Léonard Seydoux

PhD student in Seismology
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French, 27 years old

Education

2013-16 **Ph.D. in Seismology**, Institut de Physique du Globe de Paris, France
2011-13 **M.S. in Acoustics**, Université Pierre et Marie Curie, Paris 6, France
2007-11 **B.S. in Physics**, Université Paul Sabatier, Toulouse 3, France

Teaching

Total teaching hours as teacher assistant: ∼200 h.
Paris Diderot University and Institut de Physique du Globe de Paris, Paris, France

Bachelor
Python programming (20 h)
Data analysis in Earth sciences (60 h)
Experimental Physics with Python and Arduino (24 h)
Office automation (44 h)

Master
Matlab programming (44 h)
C programming practicals (12 h)

Research management

2016 (6 mth) **Master student internship supervizing:** Dr. Jean Soubestre, Institut de Physique du Globe de Paris, Laboratoire de Sismologie. *Étude de l'activité du groupe volcanique du Klyuchevskoy au Kamchatka via une méhode de réseau sismique.*
Duration: 6 months. Supervisor: Nikolaï Shapiro (Institut de Physique du Globe de Paris).

Mar. 2016 **Open Classroom at University:** *Music to understand string motion.*Paris Diderot University. Organizers: Dr. Lambert Caron and Dr. Arnaud Burtin, Institut de Physique du Globe de Paris. Duration: 1 day. Participants: ∼40 bachelor students.

Oct. 2014 **Scientific days organization:** *Seismology and tectonics on Earth: from the large scale behaviour to the impact on urban constructions.* Institut de Physique du Globe de Paris. Duration: 3 days. Visitors: ∼80 students from middle and high school, and general public.

Mar. 2014 **IPGP PhD meeting organization:** Website development, communication manager and program comity (`https://hekla.ipgp.fr/cdd2014/index.php?page=home`). Meeting duration: 1 week. Participants: 180 PhD students from IPGP + 12 external PhD students. Budget: 15000€

Peer-reviewed journals (2 published, 1 under review, 1 in preparation)

in prep. **Seydoux, L.**, Shapiro, N. M., de Rosny, J. (2016). *Anisotropic ambient noise cross-correlation: the inverse filter approach.* To be submitted.

2016 **Seydoux, L.**, Shapiro, N. M., de Rosny, J., & Landes, M. (2016). *Spatial coherence of the seismic wavefield continuously recorded by the USArray.* Geophysical Research Letters, 43. http://doi.org/10.1002/2016GL070320

2016 **Seydoux, L.**, Shapiro, N. M., de Rosny, J., Brenguier, F., & Landes, M. (2016). *Detecting seismic activity with a covariance matrix analysis of data recorded on seismic arrays.* Geophysical Journal International, 204(3), 1430–1442. http://doi.org/10.1093/gji/ggv531

2014 Le Carrou, J. L., Chadefaux, D., **Seydoux, L.**, & Fabre, B. (2014). *A low-cost high-precision measurement method of string motion.* Journal of Sound and Vibrations. http://doi.org/10.1016/j.jsv.2014.04.023

Conferences proceedings (1 talk, 2 posters)

2016 **Seydoux, L.**, de Rosny, J., & Shapiro, N. M. (2016, December). *Pre-Processing Noise Cross-Correlations with Equalizing the Network Covariance Matrix Eigen-Spectrum.* In 2016 AGU Fall Meeting (poster).

2015 **Seydoux, L.**, Shapiro, N. M., de Rosny, J., & Landes, M. (2015, December). *A Spatial Coherence Analysis of Seismic Wavefields Based on Array Covariance Matrix: Application to One Year of the USArray Data.* In 2015 AGU Fall Meeting (talk).

2014 **Seydoux, L.**, Shapiro, N., de Rosny, J., & Brenguier, F. (2014, December). *Detecting Seismic Activity with a Covariance Matrix Analysis of Data Recorded on Seismic Arrays.* In AGU Fall Meeting Abstracts (Vol.1, p.4407), (poster).

Invited seminars

2014 Invited seminar at the *Laboratoire d'Imagerie Paramétrique*, UPMC, Paris, France.

2016 Invited seminar at the *Laboratoire de Planétologie et Géodynamique*, Nantes, France.

Other presentations

2016 **Seydoux, L.**, Shapiro, N., de Rosny (2015, March). *A new approach to spatial coherence analysis of the wavefield recorded by seismic arrays : application to one year of data from the USArray.* In IPGP PhD Meeting, Paris 2016 (talk).

2015 **Seydoux, L.**, Shapiro, N., de Rosny, J., & Brenguier, F. (2015, May). *Detecting Seismic Activity with a Covariance Matrix Analysis of Data Recorded on Seismic Arrays.* In Ambient Noise Imaging and Monitoring Workshop, Cargèse 2015 (poster).

2015 **Seydoux, L.**, Shapiro, N., de Rosny, J., Brenguier, F. & Landès M. (2015, March). *Detecting Seismic Activity with a Covariance Matrix Analysis of Data Recorded on Seismic Arrays.* In IPGP PhD Meeting, Paris 2015 (poster).

2014 **Seydoux, L.**, Shapiro, N., de Rosny, J., Brenguier, F. & Landès M. (2014, March). *Detecting Seismic Activity with a Covariance Matrix Analysis of Data Recorded on Seismic Arrays.* In IPGP PhD Meeting, Paris 2014 (poster).

Skills

Programming	`Python` (package development), `MATLAB`, `C/C++`
HPC	`MPI`, `OpenMP`, `SLURM`
Geosciences	Axisem, Tau-p, Generic Mapping Tools
Operating systems	Unix (any distribution), Macintosh, Windows
Web development	HTML, CSS, PHP, SQL
Other softwares	Latex, Github, Office suite and equivalents, Adobe Creative Suite and equivalents
Languages	French (native), English (fluent), Spannish (good notions)
Other	Driving licence Autonomous diver (CMAS international level 2) Music education